INNOVATE 4 IMPACT

AI SDG GLOBAL HACKATHON 2026

Innovate today for tomorrow's impact

Team ID: I4I 249

Team Name: InnoVision

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P.S Category: Both (Software + Hardware)

P.S ID: PS-B13

Team Members

Team Leader: Name: Samruddhi Birajdar Department: Department of Electronics & Telecommunication Engineering Role: Hardware Lead	Team Member-2: Name: Manasvi Dhawale Department: Department of Computer Science and Engineering (Data Science) Role: ML Engineer / Data Pipeline
Team Member-3: Name: Swaraj Gaikwad Department: Department of Computer Science and Engineering (Data Science) Role: Backend + GNN Developer	Team Member-4: Name: Aditi Kulkarni Department: Department of Mechanical Engineering Role: System Integration & Testing

IDEA TITLE

GridSentinel - AI-Powered Rural Feeder Fault Localization with Underground Infrastructure Intelligence

The Problem:
- Rural feeders span **50-100 km** → almost **zero mid-point sensors**
- Fault location is **manual** → crews walk the entire line
- Result: **6-24 hour outages** for a fault fixable in **30 minutes**
- Root causes: conductor damage, transformer overload, vegetation contact, illegal taps, **underground corrosion**

What GridSentinel Does:
- **Detects** real-time anomalies in voltage, current & temperature
- **Localizes** fault to a specific section via **Graph Neural Network (GNN)**
- **Classifies** root cause (5 fault types) with confidence %
- **Maps** affected villages + generates **GIS crew priority route**
- **Guides** safe switching to restore partial power immediately
- **Integrates** consumer complaints as a crowdsource signal
- **TerraShield module** — detects underground tower corrosion **before** it causes faults

Innovation & Uniqueness:

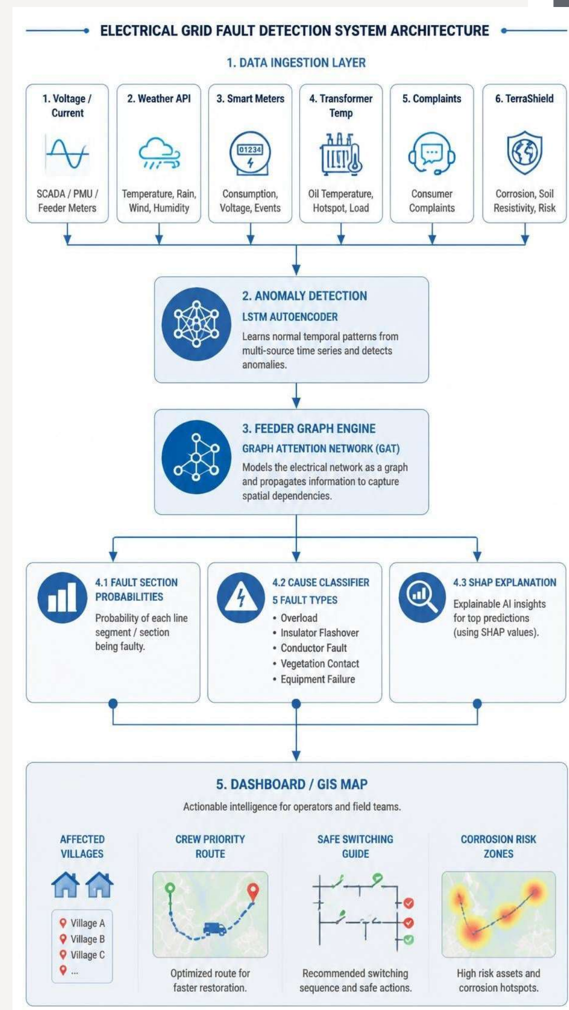
Aspect	Existing System	GridSentinel
Data Sources	1-2 Sensors	6+ (sensors + weather + complaints + underground)
Fault Logic	Fixed threshold rules	Graph Neural Network (topology aware)
Underground health	Not monitored	TFR/ERT via TerraShield
Explainability	Black Box	SHAP - “Why Section 3?”
Sparse data support	Needs Dense Sensor	Works with 1 sensor + complaints
Restoration	Stops at detection	Switching guide + crew navigation

TECHNICAL APPROACH

Tech Stack:

Layer	Technology
Anomaly Detection	PyTorch – LSTM Autoencoder
Fault Localization	PyTorch Geometric – Graph Attention Network (GAT)
Cause Classification	XGBoost / LightGBM
Explainability (XAI)	SHAP
NLP (Complaints)	spaCy + multilingual sentence-transformers
Backend API	FastAPI (Python)
Database	PgSQL + TimescaleDB
GIS / Maps	Leaflet.js
Frontend	React + Vite
Power Simulation	Pandapower
Hardware (TerraShield)	Arduino Uno + AD620 + ADS1115 + 4-electrode ERT
Deployment	Docker + Docker Compose

System Flow:



FEASIBILITY AND VIABILITY

What makes GridSentinel Feasible:

Area	Evidence
All open-source	PyTorch, FastAPI, React, Leaflet – zero licensing cost
No real grid needed	Pandapower simulates realistic rural feeders – no real grid access needed
Hardware Ready	TerraShield prototype already built (Arduino + AD620 + ADS1115)
Free data sources	IEEE PES Test Feeders + Open-Meteo API (no key needed)
Easy deployment	Docker compose – one command, runs on a single laptop

Challenges & Mitigation:

Challenge	Risk	Mitigation
No real DISCOM sensor data	High	pandapower synthetic data + IEEE feeder benchmarks
Small feeder graph	Medium	Data augmentation: fault injection at every node variant
Multilingual complaints	Medium	MiniLM supports 50+ languages incl. Hindi
Sensor Deployment Cost	High	Phased model: 3 priority sensors per feeder (head/mid/tail)
TerraShield time-scale mismatch	Medium	TFR/ERT as slow-changing background risk score (weekly)
Live hardware demo risk	Medium	Python serial mock fallback for Arduino simulation

IMPACT AND BENEFITS

Social Benefits

- Faster power restoration for schools, health centres, irrigation pumps
- Prevents crop loss for farmers dependent on electric pump irrigation
- Gives rural consumers a voice via complaint integration

Economic Benefits

- Reduces crew patrol cost (currently ₹500-₹2000/km/inspection)
- Addresses ₹1,500-₹4,000 crore/year rural economic loss from outages
- Proactive maintenance – fewer emergency replacements

Environmental Benefits

- Fewer diesel generator hours during outages (less emissions)
- Extended infrastructure life via early corrosion detection

Operational Impact:

Metric	Before	After GridSentinel
Fault search time	6-24 hours	< 1 hour
Inspection zone	50-100 km	3-5 km section
Data sources used	1-2	6+
Outage prediction	Reactive	Predictive

RESEARCH AND REFERENCES

Key References:

- IEEE PES Test Feeders: https://cmte.ieee.org/pes-testfeeders/
- pandapower: https://www.pandapower.org/
- SHAP (NeurIPS 2017): https://arxiv.org/abs/1705.07874
- Open-Meteo API: https://open-meteo.com/
- MoP RDSS Scheme 2021 | CERC Smart Grid Vision 2030

SDG Alignment:

SDG	How
SDG 7 – Affordable & Clean Energy	Reliable rural electricity access
SDG 9 – Innovation & Infrastructure	Smart grid AI for underserved areas
SDG 11 – Sustainable Communities	Resilient power for rural villages
SDG 13 – Climate Action	Reduces diesel backup + infrastructure waste

7 AFFORDABLE AND CLEAN ENERGY

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE

11 SUSTAINABLE CITIES AND COMMUNITIES

13 CLIMATE ACTION